Extra Practice 1: Zeros, Roots and x-Intercepts

A. Skills

1. Find the roots of each equation.

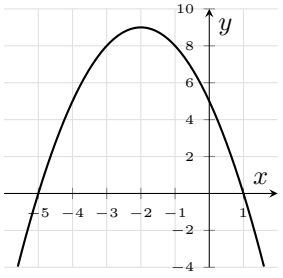
(a) $x^2 - 11x + 30 = 0$ (b) $25x^2 = 4$ (c) $2x^2 + 7x - 15 = 0$

(d) $x^3 + x^2 = 12x$ (e) $9x^2 - 30x + 25 = 0$ (f) $-x(2x + 7) = 0$

2. For each function, determine the zeros and the y-intercept of its graph.

(a) $f(x) = x^2 + 4x - 21$ (b) $g(x) = -4x^2 + 16$ (c) $h(x) = 3x^2 - 5x - 2$

3. The graph of $y = -x^2 - 4x + 5$ is shown. State the direction of opening, the vertex, the axis of symmetry, the maximum or minimum value, the zeros and the y-intercept.



B. Applying

4. Write each quadratic function in factored form.

(a) zeros -6 and 2; y-intercept -24

(b) zeros 0 and 8; passes through $(4, 12)$

5. Use a graphing calculator.

(a) Find the exact zeros of $f(x) = 12x^2 + 5x - 2$ and write $f(x)$ in factored form.

(b) Find the zeros of $y = 3x^2 - 2x - 6$ to the nearest hundredth.

6. State the number of real roots of $f(x) = 0$ for each parabola $y = f(x)$.
(a) opens up, vertex $(-3, -2)$ (b) opens down, vertex $(1, -6)$ (c) opens down, vertex $(-5, 0)$

7. A soccer ball is kicked from the ground. Its height after t seconds is $h(t) = -5t^2 + 20t$ metres.
(a) Find the zeros of h and explain what each one means.

(b) When is the ball 15 m high?

8. The x-intercepts of the graph of $y = (2x - 5)(x + 3)$ are
A. $\frac{5}{2}$ and -3 **B.** $-\frac{5}{2}$ and 3 **C.** 5 and -3 **D.** -15 only

9. Numerical Response The graph of $f(x) = a(x + 1)(x - 5)$ has a y-intercept of -20. The value of a is _____.

C. Stretch

10. The function $f(x) = 2x^2 + bx + c$ has zeros -3 and $\frac{1}{2}$. Determine b and c.

11. A parabola has x-intercepts -1 and 7, and its minimum value is -8. Write its equation in factored form.

12. One zero of $f(x) = 3x^2 + kx - 10$ is -2. Determine k and the other zero.

Extra Practice 2: Graphing $y = a(x - p)^2 + q$

A. Skills

1. Describe how the graph of each function is related to the graph of $y = x^2$.

(a) $y = x^2 + 11$

(b) $y = (x - 9)^2$

(c) $y = (x + 2)^2 - 6$

(d) $y = -4x^2$

(e) $y = 0.2(x - 1)^2 + 3$

(f) $y + 5 = -3(x + 7)^2$

2. Write the equation of the image of $y = x^2$ after the transformations, in the order given.

(a) a translation of 8 units right

(b) a translation of 2 units left and 9 units down

(c) a reflection in the x-axis, a vertical stretch by a factor of 6, and a translation of 1 unit right and 2 units up

(d) a vertical stretch by a factor of $\frac{2}{5}$ and a translation of 4 units left

3. State the vertex, the axis of symmetry and the direction of opening of each graph.

(a) $y = -(x - 4)^2 + 2$

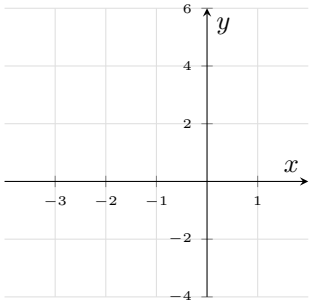
(b) $y = 3(x + 1)^2 - 7$

(c) $y - 8 = \frac{1}{2}x^2$

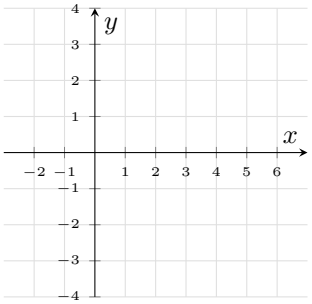
B. Applying

4. Sketch each graph using its vertex and the step pattern. State the maximum or minimum value.

(a) $y = 2(x + 1)^2 - 3$



(b) $y = -\frac{1}{4}(x - 2)^2 + 3$



5. Find the image of each point of $y = x^2$ on the given graph.
(a) $(3, 9)$ on $y = \frac{1}{3}(x + 4)^2 - 2$ (b) $(-2, 4)$ on $y = -2(x - 1)^2 + 5$

6. A translation of $y = x^2$ maps the point $(2, 4)$ to $(-1, 7)$. Describe the translation and write the equation of the image.

7. Which transformations take the graph of $y = x^2$ to the graph of $y = 4(x + 2)^2$?
A. a vertical stretch by a factor of 4 and a translation of 2 units right
B. a vertical stretch by a factor of 4 and a translation of 2 units left
C. a vertical stretch by a factor of $\frac{1}{4}$ and a translation of 2 units left
D. a translation of 4 units up and 2 units left

8. Numerical Response The graph of $y = x^2$ is stretched vertically by a factor of 3 about the x-axis and translated 5 units right and 1 unit down. The point $(3, k)$ is on the final graph. The value of k is _____.

C. Stretch

9. Use a short table of values to explain why $y = (x - 3)^2$ is 3 units to the *right* of $y = x^2$, not to the left.

10. The graph of $y = x^2$ is stretched vertically by a factor of a $(a > 0)$ and then translated 2 units left and 5 units down. The image passes through $(1, 13)$. Determine a.

11. The graph of $y = 2(x - 1)^2 + 3$ is translated 4 units left and 6 units down.
(a) Write the equation of the new graph and state its vertex.

(b) The new graph is then reflected in the x-axis. Write its equation and state its vertex.

12. The graph of $y = x^2$ is transformed to $y = a(x - p)^2 + q$. The points $(1, 1)$ and $(2, 4)$ on $y = x^2$ are mapped to $(4, 5)$ and $(5, -1)$. Determine a, p and q, and write the equation.

Extra Practice 3: Characteristics of a Quadratic Function in Vertex Form

A. Skills

1. Complete the table.

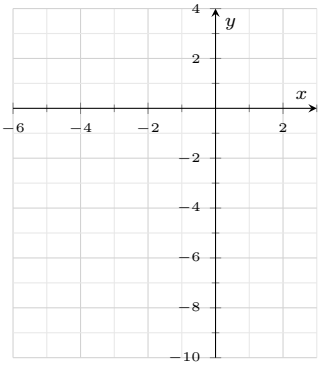
Function	Vertex	Axis	Max or min	Domain	Range
(a) $y = -4(x-1)^2 + 7$					
(b) $y = \frac{2}{3}(x+9)^2 - 1$					
(c) $y - 5 = 2x^2$					
(d) $4y = -(x-2)^2$					

2. State the y-intercept and the number of x-intercepts of each graph.
(a) $y = (x-5)^2 - 3$ (b) $y = -2(x+1)^2 - 4$ (c) $y = 3(x+2)^2$

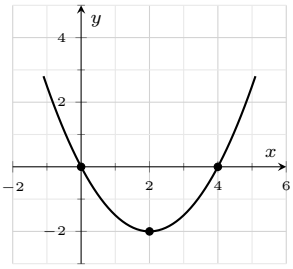
3. Determine the x-intercepts as exact values in simplest form.
(a) $y = 2(x-1)^2 - 50$ (b) $y = -(x+7)^2 + 3$ (c) $y = \frac{1}{2}(x-2)^2 - 9$ (d) $y = -3(x-4)^2 - 12$

B. Applying

4. For $y = (x+2)^2 - 9$, state the vertex, the axis of symmetry, the range, the y-intercept and the x-intercepts. Then sketch the graph.



5. The graph of a quadratic function is shown. The marked points have integer coordinates. State the vertex, the minimum value, the range, and the x- and y-intercepts.



6. A diver leaves the end of a diving board. Her height above the water is $h = -1.25(x-1)^2 + 5$ metres, where x is her horizontal distance in metres from the end of the board.

(a) What is her maximum height, and how far out does it occur?

(b) How high is the board above the water?

(c) How far from the end of the board does she enter the water?

7. The points $(-6, 4)$ and $(2, 4)$ are on the graph of $y = a(x-p)^2 + q$. Determine p. Explain.

8. Which function has a maximum value of 3?

A. $y = 3(x-2)^2$ **B.** $y = -(x-3)^2$ **C.** $y = -2(x+1)^2 + 3$ **D.** $y = (x+1)^2 + 3$

9. Numerical Response The positive x-intercept of the graph of $y = 4(x+1)^2 - 9$, to the nearest tenth, is _____.

C. Stretch

10. For which values of q does the graph of $y = -3(x+4)^2 + q$ have (a) two x-intercepts, (b) one, (c) none?

11. The graph of $y = 2(x-p)^2 + q$ has x-intercepts $3 + \sqrt{7}$ and $3 - \sqrt{7}$. Determine p and q.

12. The graph of $y = (x-p)^2 - 9$ has an x-intercept at 1. Determine the two possible values of p, and the other x-intercept in each case.

Extra Practice 4: General Form and Factored Form: Intercepts, Axis and Vertex

A. Skills

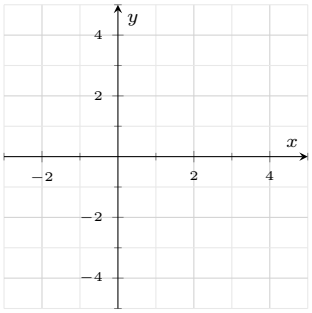
1. Write each function in general form.
(a) $y = (x - 7)(x + 2)$ (b) $y = -3(x + 1)(x + 4)$ (c) $y = 2(x + 3)^2 - 9$

2. State the zeros, the axis of symmetry, the vertex and the y-intercept.
(a) $y = (x + 8)(x - 2)$ (b) $y = -\frac{1}{2}(x - 1)(x - 9)$

3. State the axis of symmetry, the vertex, the y-intercept and the range.
(a) $y = x^2 - 10x + 21$ (b) $y = -4x^2 - 8x + 5$ (c) $y = 0.5x^2 + 4x$

B. Applying

4. Factor $y = x^2 - 2x - 3$. State the zeros, the axis of symmetry, the vertex and the y-intercept, then sketch the graph.



5. For $y = 2x^2 + 8x + 1$, find the vertex, write the function in vertex form, and determine the x-intercepts exactly and to the nearest hundredth.

6. A dolphin's leap out of the water is modelled by $h = -0.75(x-1)(x-5)$, where h is its height above the water in metres and x is the horizontal distance in metres from a buoy.

(a) Where does the dolphin leave and re-enter the water?

(b) What is its maximum height?

7. Which function has axis of symmetry $x = -2$?

A. $y = x^2 - 4x + 1$ **B.** $y = 2x^2 + 8x - 3$ **C.** $y = (x+1)(x-5)$ **D.** $y = -x^2 + 4x$

8. Numerical Response The maximum value of $y = -x^2 + 10x - 16$ is _____.

C. Stretch

9. The vertex of the graph of $y = x^2 + bx + c$ is $(3, -4)$. Determine b and c.

10. The graph of $y = 2x^2 + bx + c$ passes through $(-1, 4)$ and $(5, 4)$. Determine b and c.

11. For which values of k is the vertex of $y = x^2 + kx + 9$ on the x-axis?

12. Expand $y = a(x-r)(x-s)$ to show that its y-intercept is ars and that $x = -\frac{b}{2a}$ gives the same axis as $x = \frac{r+s}{2}$.

Extra Practice 5: Writing a Quadratic Equation from Its Vertex and a Point

A. Skills

1. Write an equation in vertex form for the parabola with the given vertex that passes through the given point.

(a) vertex $(1, 3)$, point $(2, 8)$

(b) vertex $(-6, -2)$, point $(-4, -14)$

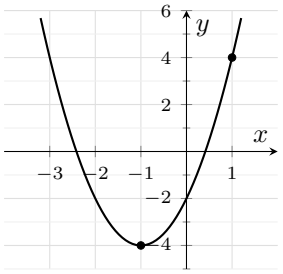
(c) vertex $(5, 0)$, point $(1, 4)$

(d) vertex $(0, 7)$, point $(-3, 1)$

2. A parabola has vertex $(-2, -5)$ and y-intercept 7. Write its equation in vertex form and in general form.

3. A parabola has zeros 1 and 7 and passes through $(5, 16)$. Write its equation in factored form and state its vertex.

4. Write the equation of the parabola shown in vertex form, then state its range and its y-intercept.



B. Applying

5. A quadratic function has vertex $(2, 9)$, and one x-intercept is 5.

(a) Determine its equation in vertex form and in general form.

(b) State the other x-intercept and the range.

6. The graph of $y = a(x + 3)^2 + q$ passes through $(-1, 2)$ and $(1, 14)$. Determine a and q, and state the minimum or maximum value.

7. The entrance to a tunnel is a parabola 8 m wide at the road and 6 m high at its centre.
(a) With the origin at the left edge of the road, determine the equation of the entrance.

(b) A truck 3 m wide and 5 m tall drives through the centre of the tunnel. Will it fit? Explain.

8. A soccer ball is kicked from the ground and follows a parabolic path. It reaches a greatest height of 9 m at a horizontal distance of 15 m. Using the kicking point as the origin, determine the equation of the path, its height at a horizontal distance of 25 m, and where it lands.

9. A parabola has vertex $(-1, 4)$ and y-intercept 1. Its equation is
A. $y = -3(x-1)^2 + 4$ **B.** $y = -3(x+1)^2 + 4$ **C.** $y = 3(x+1)^2 - 4$ **D.** $y = -(x+1)^2 + 4$

10. Numerical Response A parabola has vertex $(2, 12)$ and passes through $(0, 4)$. Its positive x-intercept, to the nearest tenth, is _____.

C. Stretch

11. A parabola passes through $(0, 5)$, $(4, 5)$ and $(6, 17)$. Determine its equation in vertex form. (Hint: find the axis first.)

12. A parabola has vertex $(1, -16)$, and its two x-intercepts are 8 units apart. Determine its equation in factored form.

13. The main cable of a suspension bridge hangs in a parabola between two towers 200 m apart. The cable is attached 50 m above the road at each tower, and its lowest point is 5 m above the road, halfway between the towers. Using the road below the lowest point as the origin, determine the equation of the cable, and the height of the cable 40 m from a tower.

Extra Practice 6: Completing the Square

A. Skills

1. Determine the number that must be added to make a perfect square, and write the result as a square. (a) $x^2 + 20x$ (b) $x^2 - 13x$ (c) $x^2 + \frac{2}{3}x$

2. Write each function in vertex form.
(a) $y = x^2 - 8x + 9$ (b) $y = x^2 + 9x + 20$ (c) $y = x^2 - 2x + 7$

3. Write each function in vertex form.
(a) $y = 3x^2 + 6x + 8$ (b) $y = -2x^2 - 12x - 11$

(c) $y = \frac{1}{3}x^2 - 2x + 1$ (d) $y = -x^2 + 5x$

B. Applying

4. Write $f(x) = 5x^2 - 20x + 26$ in vertex form. State the vertex, the axis of symmetry, the minimum value and the range.

5. Complete the square to determine the zeros of $y = x^2 + 4x - 1$, as exact values and to the nearest hundredth.

6. A diver jumps from a 10 m platform. Her height above the water, in metres, after t seconds is $h = -5t^2 + 10t + 10$. Complete the square to find her greatest height and when she reaches it.

7. A student wrote $y = -x^2 + 4x + 3 = -(x^2 + 4x + 4 - 4) + 3 = -(x + 2)^2 + 7$. Explain the error and give the correct vertex form.

8. The vertex of the graph of $y = 2x^2 + 10x + 9$ is

A. $(2.5, -3.5)$ **B.** $(-2.5, -3.5)$ **C.** $(-2.5, 2.75)$ **D.** $(-5, -16)$

9. Numerical Response The function $y = -x^2 + 6x + k$ has a maximum value of 20. The value of k is _____.

C. Stretch

10. Write $y = x^2 + 2mx + 5$ in vertex form in terms of m. For which values of m is the vertex on the x-axis?

11. Write $y = (x - 3)(x + 7)$ in vertex form by expanding and completing the square. Check the vertex using the zeros.

12. A rectangle has a perimeter of 30 m. Show that its area is $A = -w^2 + 15w$, where w is the width, and complete the square to find the greatest possible area.

13. Determine b and c so that the graph of $y = -2x^2 + bx + c$ has its vertex at $(3, 5)$.

Extra Practice 7: The Quadratic Formula

A. Skills

1. Solve with the quadratic formula. Give exact roots in simplest form.
(a) $x^2 - 4x - 1 = 0$ (b) $x^2 + 6x + 4 = 0$ (c) $2x^2 - 2x - 1 = 0$

(d) $9x^2 + 12x + 1 = 0$ (e) $x^2 - 3x + 4 = 0$ (f) $3x^2 - 10x + 5 = 0$

2. Solve. Give the roots to the nearest hundredth.
(a) $4x^2 + 7x - 3 = 0$ (b) $-x^2 + 5x + 3 = 0$

3. Solve $10x^2 - 11x - 6 = 0$ by decomposition and with the quadratic formula.

4. Solve $x^2 + 10x + 7 = 0$ by completing the square, then check with the quadratic formula.

B. Applying

5. Write in standard form, then solve. Give exact roots in simplest form.
(a) $(x - 2)(x + 5) = 2x + 1$ (b) $\frac{x}{2} + \frac{1}{x} = 2, \ x \neq 0$ (c) $0.2x^2 - 0.6x - 0.4 = 0$

6. Choose an efficient method, name it, and solve exactly.
(a) $4(x - 1)^2 = 20$ (b) $3x^2 + 7x + 2 = 0$ (c) $x^2 + 2x - 6 = 0$

7. A student solved $-x^2 + 4x + 2 = 0$ and wrote: $x = \dfrac{-4 \pm \sqrt{16 - 8}}{-2} = \dfrac{-4 \pm 2\sqrt{2}}{-2} = 2 \pm \sqrt{2}$. Find the error and give the correct roots.

8. A diver jumps from a 10 m platform. Her height above the water after t seconds is $h = -4.9t^2 + 3t + 10$ metres. When does she enter the water, to the nearest tenth of a second?

9. A rectangular patio has a perimeter of 30 m and an area of 40 m^2. Determine its exact dimensions, then round to the nearest tenth.

10. Which equation has the roots $\dfrac{3 \pm \sqrt{5}}{2}$?

A. $x^2 - 3x + 1 = 0$ **B.** $x^2 + 3x + 1 = 0$ **C.** $2x^2 - 3x + 1 = 0$ **D.** $x^2 - 3x - 1 = 0$

11. Numerical Response A ball is thrown upward with height $h = -5t^2 + 18t + 1$ metres after t seconds. The time it is in the air, to the nearest tenth of a second, is _____ s.

C. Stretch

12. The sum of the first n natural numbers is $\dfrac{n(n+1)}{2}$. Can this sum ever be exactly 500? Use the quadratic formula to explain.

13. Solve $x^2 - 4ax + a^2 = 0$ for x in terms of a, where $a > 0$. Give the answer in simplest form.

14. A rectangle is 1 unit wide and x units long ($x > 1$). Cutting a 1 by 1 square off one end leaves a smaller rectangle with the same shape, so $\dfrac{x}{1} = \dfrac{1}{x - 1}$. Determine x exactly and to the nearest thousandth. (This is the *golden ratio*.)

Extra Practice 8: The Discriminant

A. Skills

1. Determine the discriminant and the nature of the roots.
(a) $x^2 + 5x + 7 = 0$ (b) $25x^2 - 10x + 1 = 0$ (c) $2x^2 + x - 10 = 0$

(d) $x^2 - 2x - 4 = 0$ (e) $3x^2 = 4x - 2$ (f) $(x + 1)(x - 5) = 2$

2. How many x-intercepts does each graph have?
(a) $y = -x^2 + 6x - 9$ (b) $y = 4x^2 + 3x - 2$ (c) $y = -2(x - 3)^2 - 1$

3. Use the discriminant to decide which trinomial factors over the integers, and factor it. (a) $6x^2+x-12$ (b) $5x^2-3x-1$

B. Applying

4. For what values of k does $x^2 + 8x + k = 0$ have (a) two distinct real roots, (b) two equal real roots, (c) no real roots?

5. For what values of k does $kx^2 - 10x + 5 = 0$ have real roots?

6. The equation $9x^2 + kx + 4 = 0$ has two equal real roots. Determine each value of k and the root for each.

7. Show that $3x^2 + kx - 2 = 0$ has two distinct real roots for every real number k.

8. A rocket's height is $h = -5t^2 + 30t + 2$ metres after t seconds.
(a) Use the discriminant to show that it never reaches 50 m.

(b) Use the discriminant to determine its maximum height.

9. The equation $2x^2 - 4x + c = 0$ has no real roots when

A. $c < 2$ **B.** $c = 2$ **C.** $c > 2$ **D.** $c > -2$

10. Numerical Response The discriminant of $4x(x - 1) = 3 - 2x$ is _____.

C. Stretch

11. Show that $x^2 - (k + 3)x + 3k = 0$ has real roots for every value of k. Determine the roots in terms of k, and the value of k that makes them equal.

12. For what values of p does $(p + 2)x^2 - 4x + 1 = 0$ have two distinct real roots?

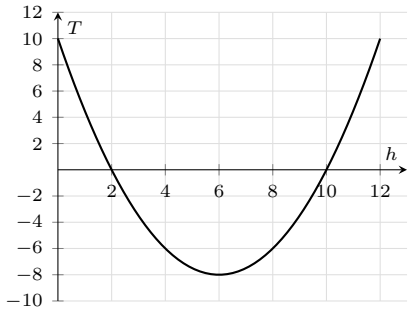
13. A rectangle has a perimeter of 20 cm. Use the discriminant to show that its area cannot be 30 cm^2, and determine the largest possible area.

14. Explain why $ax^2 + c = 0$ (no x term) has no real roots when a and c have the same sign. Give an example.

Extra Practice 9: Maximum and Minimum Problems: Graphical Approach

A. Skills

1. On a winter night, the temperature in degrees Celsius h hours after midnight is modelled by $T(h) = 0.5h^2 - 6h + 10$, $0 \le h \le 12$, graphed at the right. Use the graph.



(a) What was the temperature at midnight?

(b) What was the lowest temperature, and at what time?

(c) At what times was the temperature 0°C?

(d) State the range of the model.

2. A soccer ball is kicked. Its height in metres after t seconds is $h(t) = -4.9t^2 + 15t + 0.5$. Use a graphing calculator and answer to the nearest hundredth.

(a) What is the maximum height, and when is it reached?

(b) How high is the ball after 2 s?

(c) When does the ball land?

(d) State the domain and range in this context.

3. Name the calculator feature (value, zero, minimum, maximum or intersect) that answers each question. (a) the lowest cost (b) the height after 3 s (c) when a ball hits the ground (d) when the profit is \$500 (e) the greatest area

B. Applying

4. A bakery sells 300 cupcakes a day at \$3.00 each. For every \$0.25 increase in price, it sells 15 fewer cupcakes.

(a) Let x be the number of \$0.25 increases. Write the revenue function $R(x)$.

(b) Find the price that gives the maximum daily revenue, the number sold and the revenue.

5. A dolphin dives and comes back to the surface. Its position in metres relative to the surface (negative is below) after t seconds is $d(t) = 1.5t^2 - 9t$.

(a) How deep does the dolphin go, and when?

(b) How long is it under water?

(c) When is it 10 m below the surface? Answer to the nearest hundredth of a second.

6. A company's daily profit, in dollars, from making x bicycles is $P(x) = -15x^2 + 900x - 6000$.
(a) How many bicycles give the greatest profit, and what is it?

(b) For what whole numbers of bicycles does the company make a profit?

7. Monthly sales of a new phone, in thousands, m months after its release are modelled by $S(m) = -0.6m^2 + 9m + 4$.
(a) When do sales peak, and what are the peak monthly sales?

(b) By the model, when would sales fall to zero? Answer to the nearest tenth of a month.

8. The height of a ball is $h(t) = -5t^2 + 20t + 2$ metres after t seconds. The range of the function in this context is
A. $\{h \mid 0 \le h \le 22, h \in \mathbb{R}\}$
B. $\{h \mid h \le 22, h \in \mathbb{R}\}$
C. $\{h \mid 2 \le h \le 22, h \in \mathbb{R}\}$
D. $\{h \mid 0 \le h \le 2, h \in \mathbb{R}\}$

9. Numerical Response A golf ball's height in metres is $h(t) = -4.9t^2 + 22t$. The time the ball is in the air, in seconds to the nearest tenth, is _____.

C. Stretch

10. Two balls are thrown at the same moment. Their heights in metres are $h_1(t) = -5t^2 + 20t + 1$ and $h_2(t) = -5t^2 + 15t + 6$.
(a) Which ball goes higher, and by how much?

(b) When are the balls at the same height, and what is that height?

11. A parking lot charges \$20 a day and fills 500 spaces. For each \$1 increase, 10 fewer cars park. A city bylaw caps the price at \$28 a day.
(a) Without the cap, what price would give the maximum revenue?

(b) With the cap, what price gives the maximum revenue, and what is it? Explain.

12. A ball's height is modelled by $h(t) = -5t^2 + bt + 1.5$, and its maximum height is reached at $t = 1.6$ s. Determine b and the maximum height. (Hint: the axis of symmetry is $t = -\dfrac{b}{2a}$.)

Extra Practice 10: Maximum and Minimum Problems: Algebraic Approach

A. Skills

1. Complete the square. State the vertex and the maximum or minimum value.
(a) $y = x^2 + 8x + 5$ (b) $y = -4x^2 + 24x - 30$ (c) $y = 3x^2 - 9x + 2$

2. Determine the vertex by any algebraic method. State the maximum or minimum value.
(a) $y = -0.25x^2 + 3x - 1$ (b) $y = 0.5(x + 6)(x - 2)$ (c) $y = -5x^2 + 2x + 1$

B. Applying

3. The sum of two numbers is 22. Determine the numbers for which the sum of the first number and the square of the second is as small as possible, and that minimum.

4. A rectangular play area is built against a school wall. The other three sides use 50 m of fencing, except for a 2 m wide gate opening on the side parallel to the wall. Let each side perpendicular to the wall be x metres. Determine the dimensions that give the greatest area, and that area.

5. A pizza shop sells 160 pizzas a day at \$16 each. For every \$1 decrease in price, it sells 20 more pizzas. Determine the price for the maximum daily revenue, the number of pizzas and the revenue.

6. An arrow is shot upward. Its height in metres after t seconds is $h(t) = -5t^2 + 40t + 1.5$.
(a) Complete the square to find the maximum height and when it is reached.

(b) When does it land? Answer to the nearest hundredth of a second.

7. A rectangle has one corner at the origin, two sides along the positive axes, and the opposite corner on the line $3x + 4y = 24$. Determine the dimensions of the rectangle with the greatest area, and that area.

8. The minimum value of $y = 2x^2 - 8x + 3$ is

A. -5 **B.** 2 **C.** 3 **D.** 27

9. Numerical Response A ball's height in metres is $h(t) = -5t^2 + 24t + 1$. Its maximum height, in metres to the nearest tenth, is _____.

C. Stretch

10. At noon, ship A is at a port and ship B is 50 km due north of the port. Ship A sails east at 10 km/h and ship B sails south, toward the port, at 20 km/h. After t hours, the square of the distance between them is $d^2 = (10t)^2 + (50 - 20t)^2$. When are the ships closest, and how far apart are they then? Give the distance exactly and to the nearest tenth of a kilometre.

11. Determine k so that the maximum value of $y = -2x^2 + 12x + k$ is 25.

12. Show that, of all rectangles with perimeter P, the square has the greatest area, and that this area is $\frac{P^2}{16}$.

Extra Practice 11: Unit 5 Review — Practice Test

Calculators are allowed. Give answers exact unless the question asks you to round.

A. Multiple Choice

1. The zeros of the function $y = -2(x+4)(3x-1)$ are
A. -4 and $\frac{1}{3}$ **B.** 4 and $-\frac{1}{3}$ **C.** -4 and 3 **D.** -2, -4 and $\frac{1}{3}$

2. The graph of $y = -\frac{1}{3}(x+2)^2 + 5$ can be obtained from the graph of $y = x^2$ by several transformations. One of them is
A. a vertical stretch by a factor of 3
B. a horizontal translation 2 units left
C. a vertical translation 5 units down
D. a reflection in the y-axis

3. The graph of $y = a(x-p)^2 + q$ has **no** x-intercepts when
A. $a > 0$, $q > 0$ **B.** $a > 0$, $q < 0$ **C.** $a < 0$, $q > 0$ **D.** $a < 0$, $q = 0$

4. The range of $y = 4(x-3)^2 - 1$ is
A. $\{y \mid y \geq -1, y \in \mathbb{R}\}$ **B.** $\{y \mid y \leq -1, y \in \mathbb{R}\}$ **C.** $\{y \mid y \geq 3, y \in \mathbb{R}\}$ **D.** $\{y \mid y \in \mathbb{R}\}$

5. The vertex of the graph of $y = -2x^2 + 8x - 3$ is
A. $(2, 5)$ **B.** $(-2, -27)$ **C.** $(4, -3)$ **D.** $(2, 21)$

6. The quadratic function with vertex $(1, -8)$ whose graph passes through $(3, 0)$ is
A. $y = 2(x-1)^2 - 8$ **B.** $y = 2(x+1)^2 - 8$ **C.** $y = 4(x-1)^2 - 8$ **D.** $y = -2(x-1)^2 + 8$

7. When $y = 3x^2 + 12x + 7$ is written in the form $y = a(x-p)^2 + q$, the value of q is
A. -5 **B.** 3 **C.** 19 **D.** 11

8. The roots of $3x^2 + 5x - 1 = 0$ are
A. $\dfrac{-5 \pm \sqrt{37}}{6}$ **B.** $\dfrac{5 \pm \sqrt{37}}{6}$ **C.** $\dfrac{-5 \pm \sqrt{13}}{6}$ **D.** $\dfrac{-5 \pm \sqrt{37}}{3}$

9. The equation $x^2 - 5x + 7 = 0$ has
A. two distinct real roots **B.** two equal real roots **C.** no real roots **D.** one positive, one negative root

10. The equation $kx^2 + 4x + 1 = 0$ has two distinct real roots for
A. $k > 4$ **B.** $k < 4$, $k \neq 0$ **C.** $k = 4$ **D.** $k < -4$

B. Numerical Response

11. Numerical Response The positive root of $3x^2 = 2x + 7$, to the nearest hundredth, is _____.

12. Numerical Response The function $y = x^2 + bx + c$ has zeros 2 and 7. The value of $b + c$ is _____.

13. Numerical Response A rancher has 80 m of fencing to enclose a rectangular pen and divide it into two equal parts with a fence parallel to one pair of sides. The maximum total area, to the nearest square metre, is _____.

14. Numerical Response The product of a positive number and the number that is 3 greater is 50. The number, to the nearest tenth, is _____.

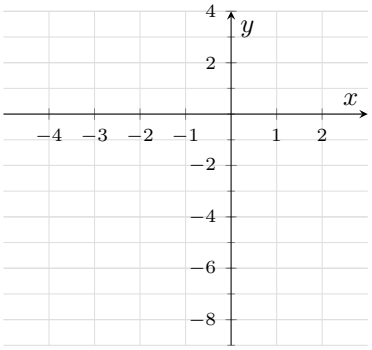
C. Written Response

15. Consider the function $y = 2x^2 + 4x - 6$.

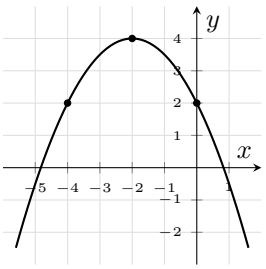
(a) Write it in vertex form by completing the square.

(b) State the vertex, the axis of symmetry, the range, the y-intercept and the x-intercepts.

(c) Sketch the graph.



16. The graph of a quadratic function is shown.



(a) Write its equation in vertex form.

(b) Determine the exact x-intercepts.

(c) Write the equation in general form.

17. (a) Use the discriminant to show that $2x^2 - 3x + 5 = 0$ has no real roots. What does this tell you about the graph of $y = 2x^2 - 3x + 5$?

(b) Determine the values of k for which $3x^2 + kx + 12 = 0$ has exactly one real root, and find the root for each value of k.

18. A water park recorded its daily attendance, A people, and the day's high temperature, T °C, all summer. A quadratic regression on a graphing calculator gave the model $A = -4.2T^2 + 260T - 2400$ for $18 \le T \le 40$.

(a) Predict the attendance on a day with a high of 22°C, to the nearest person.

(b) Which temperature, to the nearest degree, gives the greatest attendance? What is that attendance, to the nearest person?

(c) At what temperatures, to the nearest tenth of a degree, is the attendance 1400?

(d) Explain why the model should not be used for a day with a high of 10°C.

19. A bike rental shop rents 60 bikes a day at \$25 each. For every \$1 increase in price, it rents 2 fewer bikes. Write a revenue function and use it to determine the price that gives the maximum revenue, the number of bikes rented at that price, and the maximum revenue.